Multigrid Learned Primal-Dual Splitting for Artifact-Free Accelerated MRI Reconstruction

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Impact

Compact unrolled networks for accelerated MRI occasionally leave residual aliasing artifacts that undermine trust in their reconstructions. A multigrid extension suppresses these artifacts at matched size and runtime, making algorithm-derived reconstruction networks more reliable.

Synopsis

Motivation: Directly parameterized unrolled networks such as LPDSNet achieve state-of-the-art accelerated MRI reconstruction with few parameters, but can leave residual aliasing artifacts.
Goal(s): Suppress these artifacts while keeping LPDSNet's compact, algorithm-derived architecture.
Approach: We embed LPDSNet's primal-dual iterations in a multigrid V-cycle derived from MGProx. Coarse grids solve rediscretized versions of the same reconstruction problem, so spatially extended aliasing is corrected cheaply; the network is trained end-to-end.
Results: On fastMRI knee and brain at R=8 and 12, MGLPDSNet [removes the aliasing artifacts] of LPDSNet at [matched parameters and runtime], while [matching/exceeding] E2E-VarNet.

Introduction

Unrolled networks reconstruct accelerated MRI by alternating data consistency with a learned prior. Directly parameterized networks such as CDLNet[1] and LPDSNet[2] go further: every layer is one iteration of a classical algorithm, here primal-dual splitting, so the network is compact, noise-adaptive and interpretable by construction, while matching black-box networks such as E2E-VarNet[3]. In some reconstructions, however, LPDSNet leaves residual aliasing (Figure 3). Undersampling aliasing is spatially global, folding energy across the field of view, whereas each LPDS iteration applies only local filters at full resolution, so a fixed budget of iterations cannot move information far enough to undo it. Multigrid methods remove exactly this kind of spatially extended error by correcting it on coarse grids[4], and MGProx[5] extends them to the nonsmooth problems that LPDSNet solves. We propose MGLPDSNet, which embeds LPDSNet's iterations in a multigrid V-cycle.

Methods

LPDSNet: Given undersampled multicoil k-space $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x} + \boldsymbol{n}$, with $\boldsymbol{E} = \boldsymbol{M}\boldsymbol{F}\boldsymbol{S}$ (sampling mask, Fourier transform, coil sensitivities), each LPDS layer takes a primal gradient step on the image $\boldsymbol{x}$ and a dual step on latent subband coefficients $\boldsymbol{z}$ (Figure 1): learned analysis and synthesis convolutions $\boldsymbol{A}^H$, $\boldsymbol{B}$ replace the sparsifying transform, and the dual proximal map is a subband-wise clipping $\Pi_{\boldsymbol{\lambda}}$. Step sizes and thresholds are affine in the noise level σ[1,2].

MGLPDSNet: Each iteration is a V-cycle over three grid levels (N, $N/2$, $N/4$; Figure 1). LPDS layers at a level are followed by an FAS-Correction, which restricts the state to the next coarser grid and tilts that level's problem so it reproduces the fine level's residual: the full approximation scheme[4], applied to both the primal and the dual residual. Each coarse level solves the same SENSE problem posed on its own grid. Its result returns through a Coarse Correction, prolonging the coarse update with a learned step, before further LPDS layers. MGLPDSNet unrolls [6] V-cycles with [4/4/6] layers per level and $M = [81]$ subbands at every level, all weights untied.

Experiments: We trained on multicoil fastMRI[6] knee (proton density) and brain (T2-weighted) data, using measured k-space with added white Gaussian noise of $\sigma \in [0.01, 0.02]$ and 1-D random Cartesian undersampling at R=8 and 12 with a 4% fully sampled centre. Coil sensitivities were estimated with ESPIRiT from the noisy calibration region; targets were SENSE combinations of the fully sampled data. All networks were trained identically (magnitude ℓ_1+ℓ_2 loss within the anatomy mask, Adam, learning rate 5×10^{-4} with cosine decay, 3×10^5 steps). Baselines were LPDSNet[2] (30 layers, 169 subbands) and E2E-VarNet[3] given the same coil sensitivities [, and RAM[7] trained under the same protocol]. PSNR, SSIM and NRMSE were computed within the anatomy mask on 815 brain

and 100 knee test volumes at $\sigma =$ [0.015], with paired Wilcoxon signed-rank tests (Holm-corrected); network time was measured on one NVIDIA L40S.

Results

Figure 2 shows representative reconstructions for each anatomy and acceleration. [One or two sentences: where MGLPDSNet is visibly better, and what the zoomed regions show.]

Figure 3 shows LPDSNet's residual aliasing: [describe, e.g. a ghost of the anatomy shifted by FOV/R, clearly visible in the error map]. MGLPDSNet reconstructs the same slices without it. [How often it occurs: found in N of M test slices by ...]

Across 815 brain and 100 knee test volumes (Figure 4), MGLPDSNet improved on both LPDSNet and E2E-VarNet in every metric, anatomy and acceleration ($p < 0.001$), by 0.28–0.58 dB PSNR over LPDSNet and 0.28–0.57 dB over E2E-VarNet. It does so with 1.73M parameters, 17× fewer than E2E-VarNet (29.5M), at an inference time of 50 ms per slice, against 39 ms for LPDSNet and 48 ms for E2E-VarNet.

The ablations (Figure 5) show [how quality scales with the number of subbands M and of V-cycles, and the single-resolution LPDSNet row as the no-multigrid reference].

Discussion

The coarse levels give every V-cycle a path along which information travels across the image in few operations, which is what undoing global aliasing requires; local full-resolution iterations lack it. Because each coarse level solves the same reconstruction problem, the network keeps the structure of its source algorithm rather than becoming a generic multiscale CNN. [Limitations: retrospective undersampling, Cartesian sampling, a single noise model.]

Conclusion

Embedding LPDSNet in a multigrid V-cycle suppresses its residual aliasing artifacts at [matched size and runtime], while [keeping] state-of-the-art reconstruction quality.

Acknowledgements

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References

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Figures

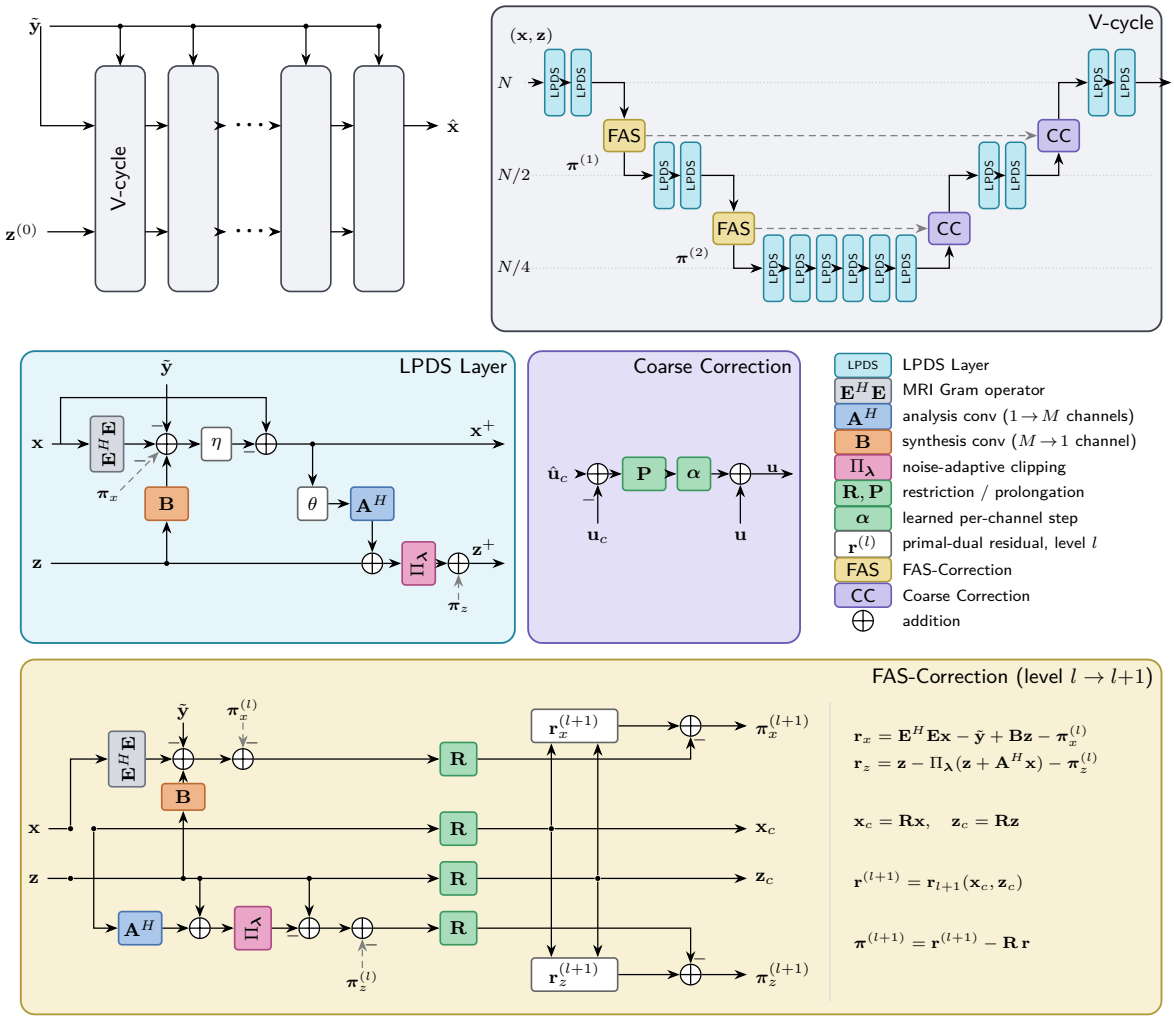


Figure 1: MGLPDSNet. Top left: [6] untied V-cycles, each receiving the preprocessed measurement $\tilde{\boldsymbol{y}} = \boldsymbol{E}^H(\boldsymbol{y} - \mu\boldsymbol{E}\mathbf{1})$, $\boldsymbol{E}$ and σ; the first layer starts from $\boldsymbol{x} = \tilde{\boldsymbol{y}}$, $\boldsymbol{z}^{(0)} = \Pi_{\boldsymbol{\lambda}}(\boldsymbol{A}^H\tilde{\boldsymbol{y}})$. Top right: one V-cycle over three grid levels; each level solves its own problem $(\tilde{\boldsymbol{y}}_l, \boldsymbol{E}_l, \sigma_l)$, tilted by the FAS correction $\boldsymbol{\pi}^{(l)}$. Middle: an LPDS Layer, with $\tilde{\boldsymbol{x}} = \boldsymbol{x}^+ + \theta(\boldsymbol{x}^+ - \boldsymbol{x})$; $\eta, \theta, \boldsymbol{\lambda}$ are affine in σ; dashed inputs (the FAS terms) exist on coarse levels only. The Coarse Correction acts on $\boldsymbol{u} \in \{\boldsymbol{x}, \boldsymbol{z}\}$ with $\hat{\boldsymbol{u}}_c$ the coarse level's output and $\boldsymbol{u}_c$ its restricted input. Bottom: the FAS-Correction, in the order its operations run; it carries its own learned copies of $\boldsymbol{A}^H$, $\boldsymbol{B}$ and $\boldsymbol{\lambda}$.

Fig. 2: reconstructions, rows knee R8 / knee R12 / brain R8 / brain R12; columns LPDSNet, E2E-VarNet, [RAM], MGLPDSNet, reference; zoom and PSNR/SSIM per panel; slices chosen by rule, not by eye.

Figure 2: [Representative reconstructions ...]

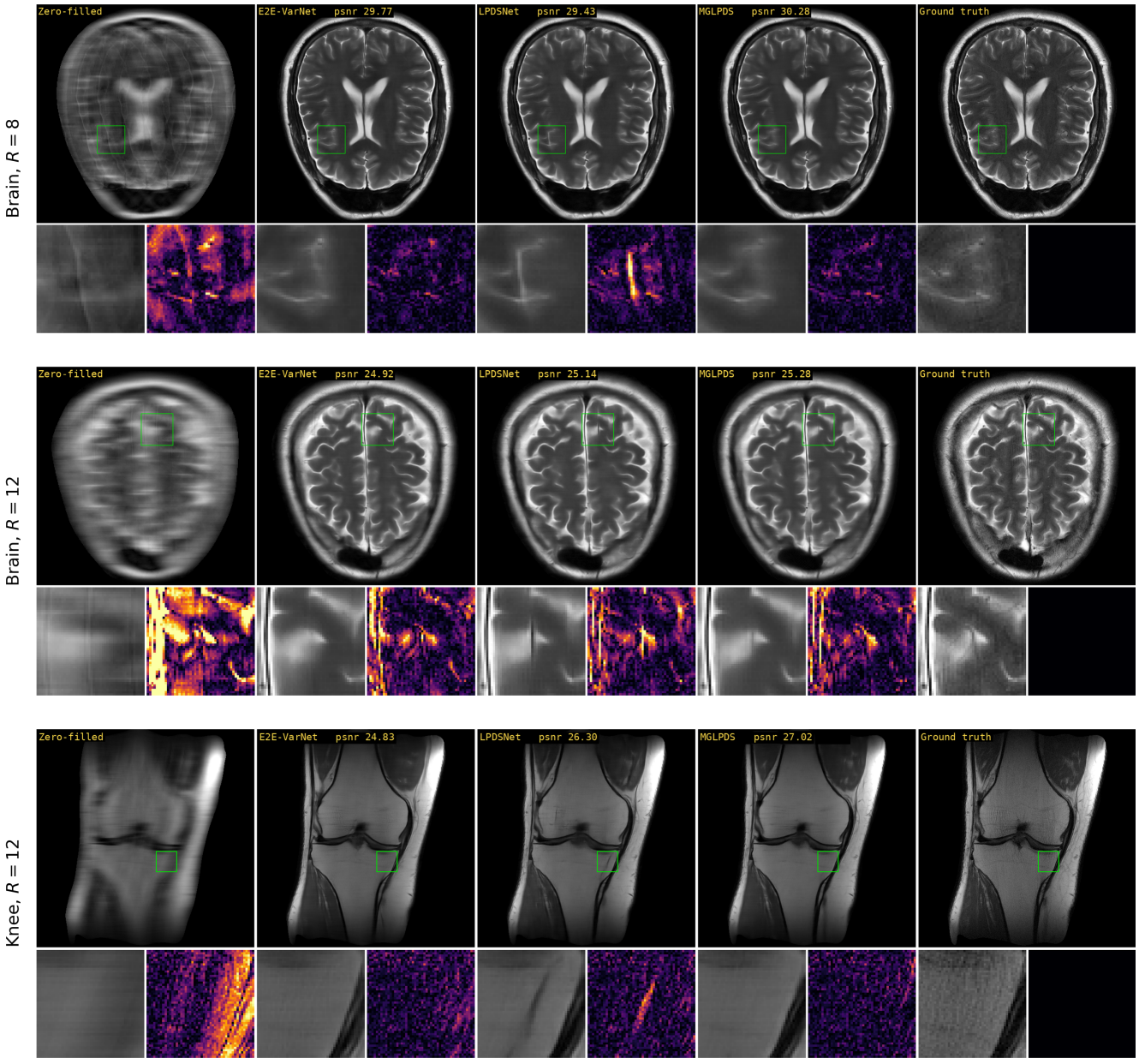


Figure 3: Residual aliasing in LPDSNet. Columns: zero-filled, E2E-VarNet, LPDSNet, MGLPDSNet and the ground truth, with PSNR per slice. Below each image: the boxed region and its absolute error. LPDSNet leaves thin line artifacts that are absent from the ground truth (brain, R=8: a bright streak; brain, R=12: a dark line; knee, R=12: an oblique streak), visible as bright lines in its error maps; MGLPDSNet and E2E-VarNet reconstruct the same regions without them.

Method	PSNR (dB) ↑	SSIM (%) ↑	NRMSE (%) ↓
Brain, R=8			
E2E-VarNet	$27.77 \pm 0.09^{*}$	$86.11 \pm 0.14^{*}$	$11.08 \pm 0.05^{*}$
LPDSNet	$27.47 \pm 0.10^{*}$	$85.06 \pm 0.16^{*}$	$11.52 \pm 0.07^{*}$
MGLPDS	$\mathbf{28.05} \pm 0.10$	$\mathbf{86.61} \pm 0.14$	$\mathbf{10.76} \pm 0.06$
Brain, R=12			
E2E-VarNet	$25.69 \pm 0.09^{*}$	$81.66 \pm 0.16^{*}$	$14.05 \pm 0.06^{*}$
LPDSNet	$25.73 \pm 0.09^{*}$	$81.48 \pm 0.17^{*}$	$14.00 \pm 0.07^{*}$
MGLPDS	$\mathbf{26.26} \pm 0.09$	$\mathbf{83.00} \pm 0.16$	$\mathbf{13.17} \pm 0.06$
Knee, R=8			
E2E-VarNet	$28.68 \pm 0.25^{*}$	$79.23 \pm 0.28^{*}$	$7.59 \pm 0.14^{*}$
LPDSNet	$28.86 \pm 0.25^{*}$	$79.12 \pm 0.28^{*}$	$7.45 \pm 0.15^{*}$
MGLPDS	$\mathbf{29.14} \pm 0.24$	$\mathbf{79.90} \pm 0.27$	$\mathbf{7.22} \pm 0.15$
Knee, R=12			
E2E-VarNet	$25.61 \pm 0.31^{*}$	$69.95 \pm 0.53^{*}$	$10.79 \pm 0.18^{*}$
LPDSNet	$25.45 \pm 0.32^{*}$	$68.88 \pm 0.54^{*}$	$10.99 \pm 0.18^{*}$
MGLPDS	$\mathbf{25.97} \pm 0.31$	$\mathbf{70.36} \pm 0.51$	$\mathbf{10.35} \pm 0.17$

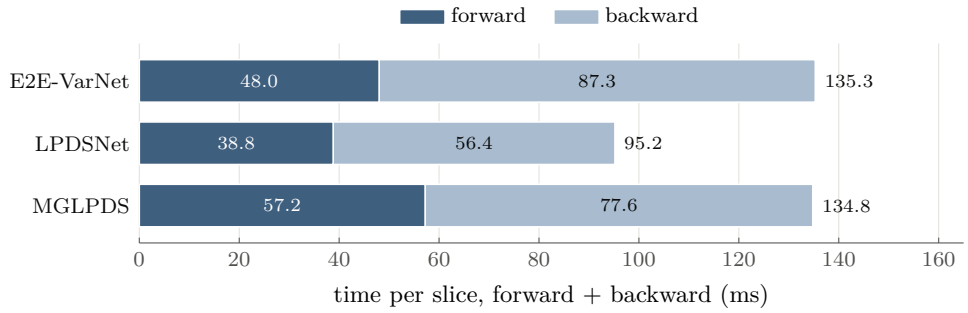
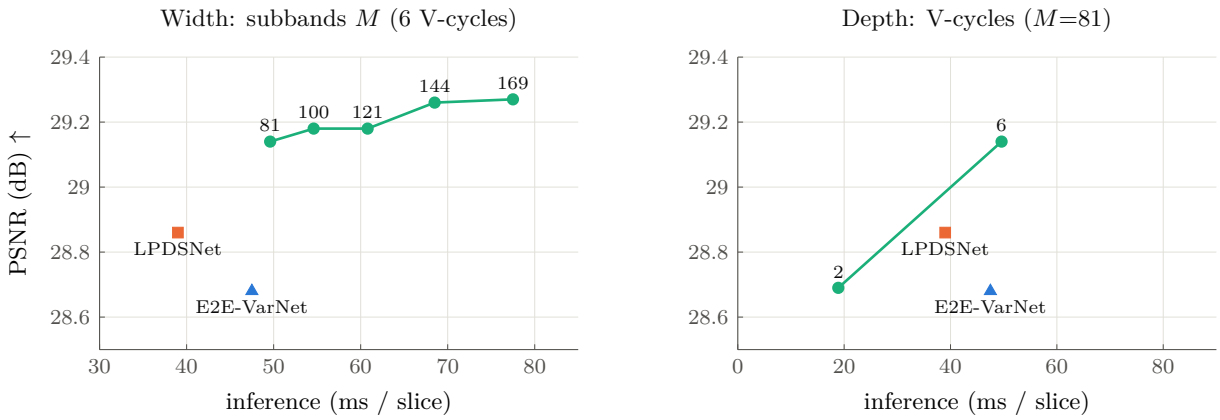


Figure 4: Quantitative comparison. Top: PSNR, SSIM and NRMSE within the anatomy mask, mean ± s.e.m. over test volumes (brain n=815, knee n=100); bold: best per experiment; *: differs from MGLPDSNet (paired Wilcoxon signed-rank over volumes, Holm-adjusted, $p < 0.05$). Bottom: network time per 640×320, 20-coil slice on one NVIDIA L40S (median of 30 runs), forward and backward pass. Parameters: E2E-VarNet 29.5M, LPDSNet 1.0M, MGLPDSNet (M=81) 1.7M.



	Params	Inference (ms)	PSNR (dB) ↑	SSIM (%) ↑	NRMSE (%) ↓
Width: subbands M, 6 V-cycles					
M=81	1.73M	49.6	29.14 ± 0.24	79.90 ± 0.27	7.22 ± 0.15
M=100	2.14M	54.6	29.18 ± 0.24*	79.93 ± 0.26	7.19 ± 0.15*
M=121	2.59M	60.8	29.18 ± 0.24*	79.99 ± 0.26*	7.19 ± 0.15*
M=144	3.08M	68.5	29.26 ± 0.24*	**80.31** ± 0.26*	7.13 ± 0.15*
M=169	3.62M	77.5	**29.27** ± 0.24*	80.25 ± 0.26*	**7.12** ± 0.15*
Depth: V-cycles, M=81					
2 V-cycles	578k	18.9	28.69 ± 0.26*	78.73 ± 0.28*	7.59 ± 0.15*
4 V-cycles	[–]	34.9	[–]	[–]	[–]
6 V-cycles	1.73M	49.6	**29.14** ± 0.24	**79.90** ± 0.27	**7.22** ± 0.15
References					
LPDSNet	1.00M	39.0	28.86 ± 0.25	79.12 ± 0.28	7.45 ± 0.15
E2E-VarNet	29.45M	47.5	28.68 ± 0.25	79.23 ± 0.28	7.59 ± 0.14

Figure 5: Ablations, knee R=8: time against quality. Top: test PSNR against inference time (640 × 320, 20 coils, NVIDIA L40S) across the width sweep (subbands M at 6 V-cycles, left) and the depth sweep (V-cycles at M=81, right), with LPDSNet and E2E-VarNet for reference. Bottom: the same networks in full, mean ± s.e.m. over n=100 test volumes; bold: best in its sweep; *: differs from M=81, 6 V-cycles (paired Wilcoxon signed-rank over volumes, Holm-adjusted, $p < 0.05$). [4 V-cycles: still training]